

PRERNA RAVI

PhD Candidate, MIT Computer Science and Artificial Intelligence Laboratory (CSAIL)

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EDUCATION

MASSACHUSETTS INSTITUTE OF TECHNOLOGY | CAMBRIDGE, MA

Ph.D., Electrical Engineering and Computer Science (EECS) | 2022 - Present

- Advisor: Hal Abelson
- Mentors: Michiel Bakker, Cynthia Breazeal, Eric Klopfer, David Karger

S.M., Electrical Engineering and Computer Science (EECS) | 2022 - 2024

- Thesis: *Large Language Model Tools for Project-Based Learning*

GEORGIA INSTITUTE OF TECHNOLOGY | ATLANTA, GA

B.S., Computer Science | 2018 - 2022

- Advisors: Thad Starner and Neha Kumar
- GPA: 4.0 / 4.0 (Highest Honors)
- Thesis: *Leveraging Sign Language Recognition in Educational Games for Deaf Children*

RESEARCH STATEMENT

My research explores how AI systems can support effective **group collaboration** and **deliberation** across diverse settings. I design frameworks and interventions that scaffold equitable team participation, foster group trust and social connection, and facilitate consensus-building. These insights inform applications in **education**, **creative practice**, and **democratic decision-making**. In parallel, I develop **AI literacy** programs for empowering stakeholders to critically engage with generative AI as both responsible creators and informed users. Situated at the intersection of **Human-Computer Interaction (HCI)** and **Artificial Intelligence (AI)**, my work employs mixed methods experiments—combining computational, qualitative and quantitative approaches—and spans participatory design, system development, and empirical evaluation.

INDUSTRY RESEARCH INTERNSHIPS

Microsoft, Research Intern | *May 2026 – Present*

Developing and evaluating socially adaptive LLM agents for culturally grounded conversational understanding; building a real-world multi-party dialogue corpus with speaker-authored implicature and face-work annotations to benchmark state-of-the-art LLMs on recovering intent, subtext, and pragmatic nuance.

Mentors: *Aaron Halfaker and Agrima Seth (Microsoft Teams Applied Sciences)*

Microsoft, Design Researcher Intern | *May 2023 – August 2023*

Co-designed generative AI tools *with and for* neurodivergent and motor disability groups within Microsoft's Windows + AI UX research team.

Google, Student Researcher Intern | *January 2022 – April 2022*

Developed new fingerspelling datasets and models for sign language recognition and integrated those into educational games for deaf children born to hearing parents. Advised by [Dr. Thad Starner](#).

Google Research Blog at I/O 2023: [Technologies for inclusive and fair ML applications](#) | [YouTube Video](#)

SOFTWARE ENGINEERING INTERNSHIPS

Microsoft, Software Engineer Intern | *May 2022 – July 2022*

Built an end-to-end Office 365 Extension for Microsoft's Artifact Management System with ML-based recommendations, used for onboarding all legal matters (involving law firms for example), their stakeholders and documents into the Office 365 Infrastructure (used by 220,000 employees).

Microsoft, Software Engineer Intern | *May 2021 – July 2021*

Developed new intelligent solutions and microservices for the Office 365 Enterprise Records Management System used for storing, migrating, and retrieving 6M+ regulatory, legal, and business-critical electronic records spanning 100 countries for 160K+ employees. Trained ML models to automatically categorize records and extract their metadata.

Microsoft, Software Engineer Intern | *May 2020 – July 2020*

Designed a centralized telemetry service and data dashboards (for web platforms used by internal consultants tracking their projects and finances) to assist debugging and product improvement, thereby directly impacting 5000+ users.

PUBLICATIONS

Pre-prints / Under Review

- [R1] Suyash Fulay*, **Prerna Ravi***, Emily Kubin, Shrestha Mohanty, Michiel Bakker, and Deb Roy. “[AI for Collective Decision-Making: Enhancing Trust and Social Cohesion.](#)” *Human-AI Complementarity Workshop: Dynamic Alignment, NSF AI Institute for Societal Decision Making (NSF AI-SDM) 2026.*
- [R2] **Prerna Ravi**, Carúmey Stevens, Brandon Hanks, Grace Lin, and Emma Anderson. “[Exploring How Agent Voice Accents Shape Human-AI Collaboration in K-12 Group Learning.](#)” *Under review: British Journal of Educational Technology (BJET) 2026*
- [R3] **Prerna Ravi***, Maria Romero*, Robert Parks, Ian Madden, Khongorzul Batireedui, Victor Lee, and Benjamin Xie. “[Building Youth Advocates' Capacity to Work with Local Environmental Data: A Design Research Study.](#)” *Under review*

Journal Papers

- [J1] **Prerna Ravi***, Dong Won Lee*, Beatriz Flámia, Jasmine David, Brandon Hanks, Cynthia Breazeal, Emma Anderson, and Grace C. Lin. “[Leveraging Large Language Models to Identify Conversation Threads in Collaborative Learning.](#)” *Journal of Educational Data Mining (jEDM) 2026.*
- [J2] Ariel Blobstein, Marc T. Facciotti, Michele Igo, David Karger, **Prerna Ravi**, Kamali Sripathi, and Kobi Gal. “[#let's-discuss: Analyzing Students' use of Emoji when interacting with course readings.](#)” *Intl. Journal of Artificial Intelligence in Education (IJAIED) 2024.*

Peer-Reviewed Conference Papers

- [C1] **Prerna Ravi**, Carúmey Stevens, Beatriz Flámia Azevedo, Jasmine David, Brandon Hanks, Hal Abelson, Grace Lin, and Emma Anderson. “[Exploring Teachers' Perspectives on Using Conversational AI Agents for Group Collaboration.](#)” *International Conference on Artificial Intelligence in Education (AIED 2026).*
- [C2] Suyash Fulay*, **Prerna Ravi***, Om Gokhale, Eugene Yi, Michiel Bakker, and Deb Roy. “[Agora: Teaching the Skill of Consensus-Finding with AI Personas Grounded in Human Voice.](#)” *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI 2026).*
- [C3] **Prerna Ravi**, John Masla, Gisella Kakoti, Grace Lin, Emma Anderson, Matt Taylor, Anastasia Ostrowski, Cynthia Breazeal, Eric Klopfer, and Hal Abelson. “[Co-designing Large Language Model Tools for Project-Based](#)

Learning with K12 Educators.” *Proceedings of the CHI Conference on Human Factors in Computing Systems (CHI 2025)*. 🏆 **Honorable Mention (Top 5%)**.

- [C4] Isabella Pu, **Prerna Ravi**, Linh Dinh, Chelsea Joe, Caitlin Ogoe, Zixuan Li, Cynthia Breazeal, and Anastasia Ostrowski. “How can we learn and use AI at the same time?: Participatory Design of GenAI with High School Students.” *Proceedings of the ACM Interaction Design and Children (IDC 2025)*.
- [C5] Grace Lin, Carúmey Stevens, Amalia Toutziaridi, **Prerna Ravi**, and Emma Anderson. “ABCDE: An Action-Oriented Framework for Collaborative Activities.” *Proceedings of the 18th International Conference on Computer-Supported Collaborative Learning (CSCL 2025)*
- [C6] Mak Ahmad, **Prerna Ravi**, David Karger, and Marc Facciotti. “How Adding Metacognitive Requirements in Support of AI Feedback Practice Exams Transforms Student Learning Behaviors” *Proceedings of the Twelfth ACM Conference on Learning at Scale (L@S 2025)* 🏆 **Best Paper Nomination**
- [C7] John Masla, Christina Bosch, **Prerna Ravi**, Lydia Guterman, Sarah Wharton, Mary Cate Gustafson-Quiett, Samar Abu Hegly, Calvin Macatantan, Eric Klopfer, Cynthia Breazeal and Hal Abelson. “Supporting AI Fluency Teaching Through the Development of Assessments for Classroom Use.” *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2025)*.
- [C8] **Prerna Ravi**, Robert Parks, John Masla, Hal Abelson, and Cynthia Breazeal. “Data comes from the real world”: A Constructionist Approach to Mainstreaming K12 Data Science Education”. *Proceedings of the ACM Virtual Global Computing Education Conference V.1 (SIGCSE Virtual 2024)*.
- [C9] Safinah Ali, **Prerna Ravi**, Katherine Moore, Hal Abelson, and Cynthia Breazeal. “A Picture is Worth a Thousand Words: Co-designing Text-to-image Generation Learning Materials for K-12 with Educators”. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2024)*.
- [C10] Safinah Ali, **Prerna Ravi**, Daniella DiPaola, Randi Williams, and Cynthia Breazeal. “Constructing Dreams using Generative AI”. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI 2024)*.
- [C11] David Kim, **Prerna Ravi**, Randi Williams, and Daeun Yoo. “App Planner: Utilizing Generative AI in K-12 Mobile App Development Education”. *Proceedings of the ACM Interaction Design and Children (IDC 2024)*.
- [C12] **Prerna Ravi**, Annalisa J. Broski, Glenda Stump, Hal Abelson, Eric Klopfer, and Cynthia Breazeal. “Understanding Teacher Perspectives and Experiences after Deployment of AI Literacy Curriculum in Middle-school Classrooms”. *Proceedings of the 16th annual International Conference of Education, Research and Innovation (ICERI 2023), IATED 2023*.
- [C13] Alex Duncan, Ana Rusch, **Prerna Ravi**, and David Joyner. “The L@St Eight Years: A Review of Papers and Authors at Learning @ Scale”. *Proceedings of the 10th ACM Conference on Learning @ Scale (L@S 2023)*.
- [C14] **Prerna Ravi**, Azra Ismail, and Neha Kumar. “The Pandemic Shift to Remote Learning under Resource Constraints”. *Proceedings of the ACM on Human-Computer Interaction (CSCW 2021)*.
- [C15] Dhruva Bansal, **Prerna Ravi**, Matthew So, Pranay Agrawal, Ishan Chadha, Ganesh Murugappan, and Colby Duke. 2021. “CopyCat: Using Sign Language Recognition to Help Deaf Children Acquire Language Skills.” *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI 2021)* 🏆 **ACM CHI Student Research Competition Winner 2021**

Short Papers and Organized Workshops

- [S1] **Prerna Ravi**, Annalisa J. Broski, Glenda Stump, Angela Daniel, Hal Abelson, Eric Klopfer, and Cynthia Breazeal. “An Art Teacher and AI: Creating Adaptable Curriculum for AI Literacy”. *Play Make Learn Conference (PML 2023)*.

- [S2] Safinah Ali, **Prerna Ravi**, Katherine Moore, Cynthia Breazeal, and Hal Abelson. “Demystifying Text-to-Image Generation for K12 Educators.” *In Workshops and Tutorials: International Society of Learning Sciences (ISLS 2023)*.
- [S3] Glenda Stump, **Prerna Ravi**, Annalisa J. Broski, Angela Daniel, Hal Abelson, Eric Klopfer, and Cynthia Breazeal. “Ethical by Design: Teaching Middle-school Students to Think Ethically About AI”. *AI Literacy Workshop at the CHI Conference on Human Factors in Computing Systems (CHI 2023)*.

White Papers

- [W1] Ben Armstrong, Julie Shah, Kate Kellogg, Sabiyyah Ali, Greer Brigham, Carey Goldberg, Shakked Noy, **Prerna Ravi**, Azfar Sulaiman, Leana Tejedor, Felix Wang, and Whitney Zhang. “Humans in the Loop: The evolution of work in early experiments with Generative AI.” *MIT Working Group on Generative AI and the Work of the Future, MIT Industrial Performance Center*.

GRANTS

- Empowering Learners with a Low-Barrier Mobile Data Science Toolkit | **Award amount: \$300,000**
 - Learning Engineering Tools Competition, 2024
 - **Prerna Ravi** (Lead researcher and proposal writer), Robert Parks, Raechel Walker, David Kim, Hal Abelson (PI)
- Agora: AI for Collective Decision Making, Enhancing Trust and Social Cohesion | **Award amount: \$70,000**
 - Foresight Institute, AI Safety Grant, 2025
 - Suyash Fulay, **Prerna Ravi** (Co-lead researcher), and Michiel Bakker (PI)

FELLOWSHIPS

- Cooperative AI Foundation PhD Fellowship, 2026
- MIT EECS Communication Fellowship, 2026
- Teaching Development Fellow, Teaching and Learning Lab (TLL) MIT, 2024-2025
- Artificial Intelligence in Education (AIED) DEIA Fellow, 2024-2025 (w/ Prof. Victor Lee, Stanford GSE)
- MIT CIS/Starr Student Travel Fellow, 2024 (UNESCO HQ speaker)
- MIT Work of the Future Fellow, 2023-2024
- Ida M. Green Memorial Fellow, 2022-2023
- MIT Vice Chancellor’s Inclusive Excellence Fellow, 2022-2023
- Adobe Research Women in Technology Scholar, 2021

AWARDS

- Special Recognition for Outstanding Reviews, CHI 2026 (2 recognitions)
- Best Paper Nomination, L@S 2025
- Best Paper Honorable Mention (top 5% of submissions), CHI 2025
- MIT Graduate Student Council (GSC) Travel Grant, 2025
- Winner - Learning Engineering Tools Competition, 2024
- Special Recognition for Outstanding Reviews, CHI 2024
- Kaufman Teaching Certificate, 2024
- Winner - ACM CHI Student Research Competition, 2021
- Georgia Tech Outstanding Junior (EDS Rising Senior) Award, 2020-2021
- Georgia Tech Outstanding Sophomore Award, 2019-2020
- Google Computer Science Research Mentorship (CSRMP), 2021

- President's Undergraduate Research Award (PURA), 2020
- Apple Women in Science and Engineering Scholarship, 2021
- Georgia Tech Faces of Inclusive Excellence Honoree, 2021
- Microsoft Invent Finalist, 2021
- Winner - Nunn School of International Affairs Paper Competition for Global Development, 2021
- Rewriting the Code Fellowship, 2020-2021
- Apple's Grace Hopper Conference Scholarship, 2020
- Honorable Mention, Microsoft Global Hackathon, 2020
- GT College of Computing Grace Hopper Conference Scholarship, 2019
- Faculty Honors for 4.0 GPA, 2018-2022

SERVICE AND NON-PROFIT WORK

- **Organizing Committee** (Inclusion and Accessibility co-chair), IUI 2027
- **Organizing Committee** (Global co-chair), CHI 2026
- **Organizing Committee** (Diversity and Inclusion co-chair), UIST 2024
- **Student Volunteer**, CHI 2025
- **Program Committee** (Senior reviewer), ISLS 2026 [4 reviews]
- **Program Committee** (Associate Chair), CHI 2025: Late Breaking Work [10 reviews]
- **Program Committee** (Senior reviewer), ISLS 2025 [5 reviews]
- **Program Committee** (Associate Chair), CSCW 2024 [4 reviews]
- **Program Committee**, MIT AI + Education Summit 2024 [10 reviews]
- **Reviewer**, CHI 2024, 2025, 2026 [4 reviews] [Special Recognition for Outstanding Reviews \(3\)](#)
- **Reviewer**, UIST 2026 [2 reviews]
- **Reviewer**, DIS 2023, 2026 [2 reviews]
- **Reviewer**, International Journal of Computer-Supported Collaborative Learning, 2025 [1 review]
- **Reviewer**, ACM Journal on Computing and Sustainable Societies, 2025 [1 review]
- **Reviewer**, Journal of Education, 2025 [1 review]
- **Reviewer**, JMIR Applications of AI, 2024 [1 review]
- **Feature Editor**, ACM XRDS Magazine, 2025 - Present
- Tech Vetter, MIT Solve Global Learning Challenge, 2024
- Workshops and Outreach, App Inventor Foundation
- Founder and President, UNICEF @ Georgia Tech, 2018-2022
- Executive Project Lead, CS + Social Good @ Georgia Tech, 2019-2022
- Training Manager, Robogals @ Georgia Tech, 2019-2020

TEACHING

Designed curricula and co-led instruction for the following courses:

Instructor **6.S062 Generative Artificial Intelligence In K-12 Education**
 Massachusetts Institute of Technology, Fall 2023.
 Enrollment: MIT & Harvard graduate and undergraduate students

Instructor **MAS.SX Text-to-Image Generation for K-12 Education**

Massachusetts Institute of Technology, IAP 2023.
Enrollment: MIT & Harvard graduate and undergraduate students, MIT staff

Instructor **Impact of Generative Artificial Intelligence within Education**
IEEE Education Society hosted by Universidad Nacional de Educación a Distancia (UNED), Fall 2023
Massive Open Online Course (MOOC)

Head Teaching **CS 1331 Introduction to Object Oriented Programming**
Assistant Georgia Institute of Technology, Spring 2019 – Fall 2021.
Enrollment: GT undergraduate students

Curriculum **Environmental Data Collection and Analysis using Micro:bits**
Developer Day of AI, 2024.
Enrollment: Middle and high school teachers & students

Instructor + **Data Science and AI with Micro:bits and MIT App Inventor**
Curriculum MIT Futuremakers Program 2024
Developer Enrollment: Middle and high school students

Instructor **Human Centered Design**
Code.X, Summer 2021.
Enrollment: Middle and high school students

MENTORING

- Jocelyn Xu, B.S. Economics, Harvard University | 2026 - *Present*
- Gisella Kakoti, B.S. Computer Science, MIT | 2024 - 2025

INVITED TALKS

- **Conversational AI Agents for Supporting and Analyzing Processes of Group Collaboration**
3rd AIED 2026 Workshop on Human-AI Teaming: "From Tools to Teammates II: Designing AI That Belongs"
- **Teaching the Skill of Consensus-Finding with AI Personas Grounded in Human Voice**
National University of Singapore (NUS) 2026
- **Co-design to Support Responsible Implementation of AI Tools in Education**
UNESCO Headquarters, France 2025
- **Designing AI Tools for Collaborative Learning Environments**
Stanford HCI Group 2025
- **Large Language Model Tools for Project-based Learning**
Lifelong Kindergarten Group, MIT Media Lab 2025
- **Democratizing K12 Data Science Education through Student-Centered Interdisciplinary Curricula**
UNESCO Headquarters, France 2024
- **The Future of Generative AI in Higher Education**
81st Annual Conference of Louisiana Colleges & Universities (CLCU) 2024
- **How Might We Redefine Learning in the Age of AI?**
Center for Constructive Communication (CCC), MIT Media Lab 2024
- **AI is for Everyone: Transforming K-12 Learning and Education in the Era of AI**
Center of Excellence in Teacher Education (CETE) at Tata Institute of Social Sciences (TISS), Mumbai, India 2023
- **PopSign: Mobile Games to teach Sign Language**
Imagine RIT 2022, Rochester, NY